

9 (long) The function $y(x)$ in Fig. 1 is odd and has period $2L$.

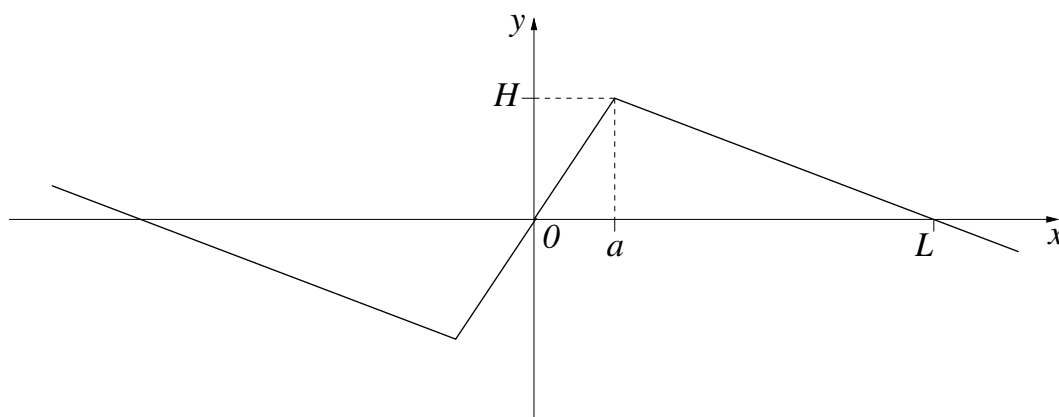


Fig. 1

(a) Show that

$$y(x) = \sum_{n=1}^{\infty} \frac{2HL^2}{n^2\pi^2a(L-a)} \sin\left(\frac{n\pi a}{L}\right) \sin\left(\frac{n\pi x}{L}\right)$$

is a Fourier series representation of $y(x)$.

[15]

(b) Find a Fourier series for $y(x)$ in the limit as $a \rightarrow 0$. You should either take the limit of the expression in (a), or manipulate the Fourier series for the saw-tooth wave in the *Mathematics Data Book*.

[10]

(c) Comment on the rates of convergence of the series in (a) and (b).

[5]